Instructor & Course Manager Dashboard Features

Complete Feature Explanation Guide

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NELC-Compliant eLearning Platform • Bilingual AR/EN • WCAG 2.1 AA

Milestone M4 — Instructor + Course Manager Dashboard Development

Scope	Two complete dashboards — Instructor (11 pages, 27 components) + Course Manager (8 pages, 20+ components)
Audience	Developers, QA testers, product owners, stakeholders, NELC auditors
Milestone	M4 — Full implementation with Zoom API, attendance tracking, NELC audit logs
NELC Rules	Virtual class limits (≤40 long / ≤50 short), auto assistant allocation, attendance sync, audit evidence export
Authorization	INSTRUCTOR role: own data only | COURSE_MANAGER role: all courses, instructors, content oversight

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PART A

Instructor Dashboard

Plan and deliver sessions, monitor learner progress, and continuously improve course quality

Role: INSTRUCTOR | Pages: 11 | Components: 27 | Authorization: Own courses and students only | Data isolation: WHERE instructorId = currentUser.id

1. Instructor Dashboard Home — The Command Center

Page: (dashboards)/dashboard/instructor/page.tsx

The dashboard home is the first screen instructors see after login. It provides a real-time operational overview of everything that matters today — upcoming sessions, student performance, at-risk learners, and pending certificate reviews. Think of it as a morning briefing for the instructor.

1.1 KPI Cards (Top Row)

Four metric cards displayed across the top of the dashboard, giving an instant pulse check:

Card	What It Shows	Data Source	Why It Matters
Active Courses	Count of courses with status='active'	GET /courses (filtered)	Shows current teaching load
Today's Sessions	Count of sessions scheduled for today	GET /sessions/today	Immediate action items
Total Students	Unique enrolled students across all courses	Aggregated from enrollments	Reach metric
At-Risk Learners	Students flagged by at-risk criteria	GET /instructor/at-risk	Urgency indicator — needs attention

1.2 Today's Sessions Widget

Shows all sessions scheduled for the current day, ordered by time. Each session card displays:

- Session title and course name
- Scheduled time and duration
- Delivery type indicator: Zoom icon (online), building icon (on-site), or hybrid icon
- Expected attendance count vs enrolled students
- Zoom link or room/venue information
- **One-Click Start button** — changes session status from 'scheduled' to 'live', records the actual start time

User Flow: Instructor arrives at 9am → sees 'Advanced Physical Therapy — 9:30 AM' → clicks 'Start' at 9:28 → status changes to LIVE → Zoom link copied to clipboard → attendance tracking begins.

1.3 Course Performance Widget

A chart-based widget showing enrollment trends and completion analytics for the instructor's active courses:

- 7-day enrollment trend line chart

- Completion rate per course as horizontal progress bars
- Drop-off analysis showing where students exit (which module/video)
- Comparison: this week vs last week

1.4 Assessment Analysis Widget

Summarizes assessment performance across all active courses:

- Average scores per assessment/module
- Pre-test vs post-test delta (measuring learning improvement)
- Quick stat: number of submissions pending grading

1.5 At-Risk Learners Widget

Critical feature for NELC compliance — the platform must proactively identify and support struggling students.

Displays students who meet ANY of these criteria:

Risk Type	Criteria	Badge Color	Suggested Action
Low Attendance	Attendance < 70% across all sessions	Red	Send reminder email / schedule make-up session
Low Score	Assessment average < 60%	Orange	Offer additional resources / review session
Inactive	No platform activity in last 7 days	Yellow	Send engagement email / check-in

Each student card shows: name, photo, enrolled course, specific risk type, and action buttons (Send Reminder, Schedule Make-up, Open Ticket).

1.6 Certificate Queue Widget

Shows pending certificate requests that need instructor review. Displays: student name, course, eligibility status (attendance %, assessments passed, course completed), and approve/reject buttons. Badge count indicates how many are waiting.

1.7 Calendar & Notifications

Mini monthly calendar with session dots color-coded: blue = upcoming, green = completed, red = cancelled. Notifications widget shows platform alerts (new enrollment, session reminder, certificate approved).

2. My Courses — Course Portfolio

Page: instructor/courses/page.tsx

A grid/list view of all courses owned by the instructor. This is the entry point for course management.

2.1 Course Cards

Each course displays as a visual card with:

- Thumbnail image (uploaded or auto-generated)
- Course title (in current language — AR or EN)
- Delivery type badge: Zoom / On-site / Blended
- Status badge: Draft (gray), Active (green), Completed (blue), Archived (dark)
- Quick stats: enrolled students count, completion rate, average rating
- Date range (start date – end date)

2.2 Filters & Search

- Status filter: Draft / Active / Completed / All
- Delivery type filter: Zoom / On-site / Blended / All
- Search bar: real-time search by course title (debounced 300ms)
- Sort: Newest first, Oldest first, Most enrolled, Alphabetical
- View toggle: Grid view (cards) / List view (table)

2.3 Create Course Flow

The 'Create Course' button opens a multi-step form:

Step	Fields	Validation
Basic Info	Title (EN), Title (AR), Description (EN/AR), Delivery type, Thumbnail upload	Title required, delivery type required
Schedule	Start date, End date, Max enrollment	End date > start date
Modules	Add/reorder modules: title, type (video/doc/quiz), estimated duration	At least 1 module recommended
Review	Preview all settings before publishing	Saves as 'draft' by default

3. Course Detail — 6-Tab Deep Dive

Page: instructor/courses/[courseId]/page.tsx

When an instructor clicks into a course, they see a detailed view organized into 6 tabs. Data loads lazily — only when a tab is selected.

3.1 Overview Tab

Course summary: title, description, delivery type, status, date range, enrollment count, completion rate, average rating. Edit button opens inline editing.

3.2 Modules Tab

Ordered list of course modules with drag-and-drop reordering. Each module shows: title, type icon, estimated duration, material count. Actions: add, edit, delete, reorder.

3.3 Students Tab

Enrolled students table: name, email, enrollment date, progress bar (0-100%), status, last activity. Click navigates to student detail page.

3.4 Sessions Tab

All sessions for this course. Shows date, time, type, status, attendance count. Quick actions: schedule, start, view attendance.

3.5 Analytics Tab

- Enrollment funnel: views → enrolled → active → completed
- Completion rate and drop-off by module
- Average progress and attendance rate
- Assessment performance: average scores, pass/fail rates

3.6 Settings Tab

Edit course details, status management (activate, complete, archive), and danger zone (delete drafts only).

4. Sessions — Schedule, Deliver & Track

Page: instructor/sessions/page.tsx

4.1 List & Calendar Views

Sessions displayed in two views: List (grouped by Today/Upcoming/Past) and Calendar (monthly with color-coded dots). Filters: course, status, date range.

4.2 Schedule Session Modal

- Course selection dropdown (from instructor's courses)
- Session title, date/time picker, duration (30/60/90/120 min)
- Type: Zoom / On-site / Hybrid
- If Zoom: paste meeting link or auto-generate (Phase 2)
- If On-site: room name + venue address
- Conflict detection: warns if another session overlaps

4.3 Session Lifecycle

Status	Trigger	What Happens
Scheduled	On creation	Appears in calendar, students see upcoming session
Live	Instructor clicks 'Start'	actualStartedAt = NOW(), attendance tracking begins
Completed	Instructor clicks 'End'	actualEndedAt = NOW(), attendance finalized
Cancelled	Instructor cancels	cancelReason stored, students notified

4.4 Zoom Integration (NELC-Compliant)

NELC Requirement: Virtual class management with specific learner limits and automated attendance.

- Zoom API (Webhook-based) integration for live sessions
- Real-time minute-by-minute attendance tracking via Zoom webhooks
- **Learner limits enforced:** ≤40 for long courses, ≤50 for short courses (NELC regulation)
- Auto assistant allocation when class exceeds threshold
- Attendance sync: Zoom join/leave events automatically recorded in attendance table
- Automated NELC audit logs generated for every session

4.5 Attendance Tracking

- Bulk mark: select all students, mark as present/absent/late/excused in one click
- Individual mark: click each student's status toggle
- Late/excused requires a note (reason)

- Minute-level attendance records: joinedAt, leftAt, durationMinutes
- NELC evidence export: detailed attendance report with timestamps

5. Students — Monitor & Support

Pages: instructor/students/page.tsx + instructor/students/[studentId]/page.tsx

5.1 Students List

Aggregated view of ALL students across the instructor's courses. Table: name, email, courses count, progress, attendance rate, assessment average, risk status, last active. Filters: search, course, status, risk level.

5.2 Student Detail (4 Tabs)

Tab	Content	Key Data
Overview	Profile + enrollment summary across all courses	Total courses, avg progress, avg attendance, avg score
Attendance	Session-by-session records for each course	Status per session, total rate, late/absent counts
Assessments	Scores across all courses	Each score, pass/fail, grade vs class average
Activity	Timeline of student actions (future)	Last login, engagement score

6. Assessments — Evaluate & Analyze

Page: instructor/assessments/page.tsx

6.1 All Assessments Tab

List of all assessments across courses. Each shows: title, course, type (quiz/exam/assignment), marks, status (draft/published/closed), submission count. Filter by type, status, course.

6.2 Question Bank Tab

Key concept: Questions can belong to a specific assessment OR exist in the course-level bank (assessmentId = null). Bank questions can be pulled into any assessment.

- View all questions organized by topic/difficulty
- Filter by type (MCQ, True/False, Short Answer, Essay), difficulty (Easy/Medium/Hard)
- Create: question text (EN + AR), options for MCQ, correct answer, marks, difficulty

6.3 Submissions & Grading

- Auto-graded: MCQ and True/False scored automatically on submission
- Manual grading: Essay and Short Answer require instructor review
- Grade modal: student answers side-by-side with correct answers

6.4 Item Analysis (NELC Required)

Metric	Formula	Interpretation
Difficulty Index	correct_answers / total_attempts	0.0=impossible, 1.0=trivial. Ideal: 0.3–0.7
Discrimination Index	top_27%_correct - bottom_27%_correct	<0.2=poor, 0.2–0.4=good, >0.4=excellent
Pre/Post Comparison	post_test_avg - pre_test_avg	Measures actual learning improvement

7. Content Management — Upload & Version

Page: instructor/content/page.tsx

7.1 Material Library

Grid/list view of all materials. Each card: thumbnail/icon, title, course, file size, version number, processing status (Processing/Live/Failed), WCAG badge (Pending/Pass/Warning/Fail).

7.2 Upload & Versioning

- Upload via multipart/form-data, file validation (100MB video, 10MB docs)
- Replace creates new version (v1 → v2), old version preserved via previousVersionId chain
- Captions: upload AR/EN caption files (.srt/.vtt) for WCAG compliance
- Transcripts: upload AR/EN transcripts for accessibility

8. Certificates — Review & Issue

Page: instructor/certificates/page.tsx

8.1 Eligibility (Auto-Computed)

Check	Computation	Threshold
Attendance	count(present+late) / count(all_sessions) × 100	Must reach 70% (NELC requirement)
Assessments	Did student pass ALL required assessments?	Score ≥ passing marks on each
Completion	enrollment.progress === 100?	Must be 100%

8.2 Review Actions

- Approve: certificate PDF generated, student notified, certificateUrl stored
- Reject: reason stored in reviewNote, student sees why
- Hold: audit note stored for edge cases requiring investigation
- Batch operations: select multiple, batch approve/reject

9. Reports & NELC Evidence Export

Page: instructor/reports/page.tsx

Report	Content	NELC Use
Attendance	Student name, each session date/status, overall %	Mandatory NELC evidence
Gradebook	Each assessment score, total, pass/fail	Academic records
Enrollment	Date, status, progress, last activity	Engagement tracking
Completion	Completed students, date, certificate status	Program metrics

All reports: select course → set date range → choose format (CSV/PDF/Excel) → generate → download. Every export includes timestamp, filters, and instructor name for audit traceability.

10. Instructor Settings

- Profile: name (EN/AR), photo, contact email, phone, bio/specialty
- Notifications: toggle email alerts for enrollments, sessions, certificates, at-risk
- Language: Arabic/English toggle, font size, high contrast, default captions

PART B

Course Manager Dashboard

Coordinate courses end-to-end, supervise materials and instructors, and ensure NELC compliance

Role: COURSE_MANAGER | Pages: 8 | Authorization: All courses, all instructors, content review | Scope: Platform-wide oversight (not restricted to own data)

11. Course Manager Dashboard Home

Page: (dashboards)/dashboard/course-manager/page.tsx

The Course Manager sees the big picture across ALL courses and instructors. Unlike the Instructor dashboard (which is personal), this dashboard aggregates data platform-wide for quality oversight and NELC compliance.

11.1 KPI Cards (Top Row)

Card	What It Shows	Data Source	Why It Matters
Total Courses	Count by status: Active/Preparing/Completed	GET /cm/courses/stats	Portfolio health at a glance
Total Enrollments	All enrollments this month + trend	Aggregated enrollments	Growth tracking
Avg Completion Rate	Average across all active courses	Computed from enrollments	Quality metric
Pending Reviews	Content items awaiting approval	GET /cm/content-reviews?status=pending	Bottleneck indicator
Active Instructors	Instructors currently teaching	GET /cm/instructors/active	Capacity planning
NELC Score	Overall compliance percentage	GET /cm/compliance/score	Audit readiness

11.2 Catalog Overview Widget

Visual breakdown of all courses across the platform:

- Course state distribution: donut chart showing Active (green), Preparing/Draft (yellow), Completed (blue), Archived (gray)
- Enrollment counts per course as horizontal bars, ranked by popularity
- Completion rate heatmap: courses color-coded from red (<50%) to green (>80%)
- Course rating averages: star display with student count
- Click any course to drill into its detail view

11.3 Content Review Queue Widget

This is a KEY differentiator from Instructor Dashboard — Course Managers approve content BEFORE it goes live.

Shows all content items submitted by instructors that need pre-publish review:

Check	What Is Verified	Pass Criteria
Learning Objectives	Are objectives clearly defined?	At least 3 measurable objectives per module

Assessment Alignment	Do assessments match objectives?	Each objective covered by at least 1 assessment
Hours Compliance	Does content meet NELC required hours?	Course hours ≥ NELC minimum for category
WCAG Compliance	Are materials accessible?	Captions present, alt text on images, contrast ratio pass
NELC Checklist	Platform-specific NELC requirements met?	All mandatory fields filled, policies linked

Each item in the queue shows: material title, course name, instructor who submitted, date submitted, checklist status (pass/fail per check), and action buttons: Approve, Reject with feedback, Request Changes.

12. Schedules & Sessions Management

Page: course-manager/sessions/page.tsx

While instructors manage their own sessions, the Course Manager has oversight of ALL sessions across the platform.

12.1 Cross-Instructor Schedule View

- Calendar view showing ALL sessions from ALL instructors in one view
- Color-coded by instructor for easy identification
- Conflict detection: highlights when two instructors are scheduled for overlapping times
- Capacity planning: seats available vs enrolled students per session
- Filter by instructor, course, delivery type, date range, venue

12.2 Capacity Planning

Metric	Calculation	Action If Exceeded
Zoom Sessions	Enrolled vs Zoom license capacity	Warn CM, suggest split into multiple sessions
On-site Sessions	Enrolled vs room max capacity	Block enrollment beyond room limit
NELC Limits	≤40 for long courses, ≤50 for short	Auto-enforce: prevent enrollment, alert CM
Instructor Load	Sessions per week per instructor	Warn if >15 hours/week, suggest redistribution

12.3 Session Oversight Actions

- View any session's attendance record (read-only — only instructor marks attendance)
- Cancel sessions on behalf of absent instructors (with reason)
- Reschedule sessions by adjusting date/time and notifying enrolled students
- Assign substitute instructors when original is unavailable

13. Instructors Management

Page: course-manager/instructors/page.tsx

Unique to Course Manager role — instructors do NOT have this view.

13.1 Instructor Roster

Table of all instructors on the platform:

- Name, photo, specialty/department, contact info
- Active courses count, total students, sessions this week
- Teaching load bar: visual indicator of hours/week (green <10h, yellow 10-15h, red >15h)
- Quality KPIs: average course rating, student satisfaction score, completion rate
- Status: Available / Busy / On Leave

13.2 Assign Instructors to Courses

The Course Manager can:

- Assign an instructor to a course that has no instructor yet
- Reassign a course from one instructor to another (e.g., when instructor goes on leave)
- View instructor availability calendar before assigning
- Balance load: system suggests the least-loaded qualified instructor

13.3 Instructor Performance Dashboard

KPI	Source	Benchmark
Average Rating	Student feedback surveys	Target: ≥4.0 / 5.0
Completion Rate	Students who finished vs enrolled	Target: ≥75%
Attendance Rate	Student attendance in instructor's sessions	Target: ≥80%
Content Quality	WCAG pass rate of submitted materials	Target: 100% pass
Responsiveness	Time to respond to student questions	Target: <24 hours

14. Assessments Oversight

Page: course-manager/assessments/page.tsx

The Course Manager reviews assessment quality across all courses, ensuring consistency, fairness, and NELC compliance.

14.1 Blueprint Coverage

What is a blueprint? A mapping of assessment questions to learning objectives, ensuring every objective is tested.

- View assessment blueprint: which questions map to which learning objectives
- Coverage analysis: identify objectives with no assessment questions (gaps)
- Difficulty distribution: percentage of easy/medium/hard questions should be balanced
- Flag assessments with poor coverage for instructor revision

14.2 Proctoring Configuration

- Configure proctoring rules per assessment type:
  - Quizzes: no proctoring (honor system)
  - Exams: browser lockdown, webcam monitoring, or in-person proctoring
  - Assignments: plagiarism check integration (future)
- Time limits and attempt restrictions set at course manager level
- Emergency override: CM can extend time or grant extra attempts

14.3 Retake Rules

Rule	Default	CM Can Override
Max Retakes	1 retake per assessment	Yes — set per course or per student
Cooling Period	48 hours between attempts	Yes — reduce to 24h or remove
Score Policy	Highest score kept	Yes — can switch to latest score or average
Retake Scope	Full assessment	Yes — can allow partial retake (failed sections only)

15. NELC Compliance Center

Page: course-manager/compliance/page.tsx

This is the most critical Course Manager feature — ensuring every course meets NELC standards for health sector eLearning.

15.1 Compliance Dashboard

A real-time compliance scorecard for every course:

Category	Checks	Status Colors
Course Structure	Has objectives, correct hours, proper modules, bilingual content	Green=pass, Red=fail, Yellow=partial
Accessibility	WCAG 2.1 AA: captions, alt text, contrast, keyboard navigation	Green=100%, Yellow=partial, Red=<80%
Virtual Classes	Learner limits (≤40/50), assistant allocation, attendance sync	Green=compliant, Red=violation
Assessment Quality	Blueprint coverage, item analysis done, pre/post tests exist	Green=all pass, Yellow=review needed
Policies	Academic integrity, IP policy, accessibility statement published	Green=all present, Red=missing
Data Protection	PDP compliance, data retention, privacy policy, consent	Green=compliant, Red=gaps

15.2 NELC Evidence Export

- Generate compliance evidence bundles for NELC auditors
- Each bundle includes: course details, instructor credentials, content audit, assessment analysis, attendance records, student feedback, accessibility report
- Export as ZIP file containing: PDF reports, CSV data, screenshot evidence, policy documents
- Mapped directly to NELC clause numbers for faster audit readiness

15.3 Virtual Class Compliance (M4 Specific)

NELC Implementing Rules: Specific regulations for virtual classrooms in health sector training.

Rule	Requirement	Platform Enforcement
Learner Limit (Long)	Maximum 40 learners per virtual class for courses >8 hours	Auto-block enrollment at 40, alert CM
Learner Limit (Short)	Maximum 50 learners per virtual class for courses ≤8 hours	Auto-block enrollment at 50, alert CM
Assistant Allocation	Assign teaching assistant when class exceeds 25 learners	Auto-suggest TA assignment, CM approves

Attendance Sync	Minute-by-minute attendance must be recorded automatically	Zoom webhook captures join/leave events
Session Recording	All virtual sessions must be recorded and archived	Auto-recording enabled, linked to session
Audit Logs	Complete audit trail for every session	Immutable logs: start, end, attendance, actions

16. Course Manager Reports (NELC/KPIs)

Page: course-manager/reports/page.tsx

Course Manager reports are platform-wide (not course-specific like instructor reports). They serve two purposes: internal quality monitoring and external NELC audit evidence.

16.1 Report Types

Report	Scope	Audience	Content
Enrollment Summary	All courses	Admin, CM	Enrollment trends, conversion rates, demographics
Completion Analysis	All courses	Admin, NELC	Completion rates by course/category, drop-off analysis
Satisfaction Survey	All courses	Admin, NELC	Student ratings, NPS scores, verbatim feedback
Instructor Performanc e	All instructors	Admin, CM	Ratings, load, completion rates, content quality
NELC Compliance Matrix	Platform-wide	NELC Auditors	Clause-by-clause status with evidence links
Attendance Compliance	All sessions	NELC Auditors	Attendance rates vs NELC minimums per course

16.2 Export Capabilities

- All reports exportable as CSV, PDF, and Excel
- Every export includes: timestamp, date range, filters applied, generated by (user name + role)
- Scheduled reports: configure weekly/monthly auto-generation and email delivery to admin
- Evidence bundles: one-click export of all NELC-required evidence for a specific course

17. Course Manager Settings

Page: course-manager/settings/page.tsx

- Profile: name (EN/AR), photo, department, contact info
- Notifications: alerts for content review submissions, compliance violations, instructor load warnings
- Default policies: set default retake rules, proctoring settings, NELC checklist templates
- Language & accessibility: Arabic/English toggle, font size, high contrast

PART C

Comparison & Cross-Cutting Features

How the two dashboards relate and what they share

18. Instructor vs Course Manager — Role Comparison

Both roles interact with courses and students, but from fundamentally different perspectives:

Feature	Instructor	Course Manager
Scope	Own courses only	All courses platform-wide
Courses	Create, edit, deliver	Oversee, review, approve, assign instructors
Sessions	Schedule & deliver own	View all, resolve conflicts, reassign
Students	Own students only	All students across all instructors
Content	Upload materials	Review & approve before publish
Assessments	Create & grade	Oversight: blueprint, proctoring, retake rules
Certificates	Review & approve/reject	Not directly involved (instructor handles)
Reports	Per-course exports	Platform-wide analytics & NELC evidence
Compliance	WCAG on own content	Full NELC compliance monitoring
Instructors	N/A	Manage roster, assign, monitor KPIs
Data Access	WHERE instructorId = me	All data (read) + assignment powers (write)

19. Cross-Cutting Features (Both Dashboards)

19.1 Bilingual Support (AR/EN)

- Every page supports full Arabic and English rendering
- RTL layout automatically applied when language is Arabic
- All content fields have dual inputs: title (EN) + titleAr (AR)
- Language toggle in the top navigation bar, persists across session

19.2 Global Filters

- Time range: Today / 7 days / 30 days / Custom
- Course filter: specific course or all
- Delivery mode: Zoom / On-site / Blended
- Instructor filter (Course Manager only): filter by specific instructor

19.3 Export Everywhere (NELC Evidence)

- Every data table and widget supports CSV/PDF export
- Export includes: timestamp, active filters, user context, role
- Satisfies NELC evidence requirements for health sector audits

19.4 Authorization & Data Isolation

Rule	Instructor	Course Manager
JWT Token	Required on every request	Required on every request
Role Guard	@Roles('INSTRUCTOR')	@Roles('COURSE_MANAGER')
Data Filter	WHERE instructorId = currentUser.id	No filter (sees all courses)
Write Access	Own courses/sessions/materials only	Assign instructors, approve content, set policies
Admin Fallback	Admin/SuperAdmin can see all	Admin/SuperAdmin can see all

19.5 Responsive Design

- All pages work on desktop (1200px+), tablet (768–1199px), and mobile (<768px)
- Sidebar collapses to hamburger menu on mobile
- Data tables become scrollable cards on mobile
- Touch-friendly: buttons minimum 44px tap target

19.6 Loading, Error & Empty States

State	Visual	When
Loading	Skeleton placeholder (gray shimmering blocks)	While API request is in flight
Error	Red error banner with 'Retry' button	API returns error or network failure
Empty	Illustration + helpful message + action button	No data exists yet

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